

Information Security : TP Privacy

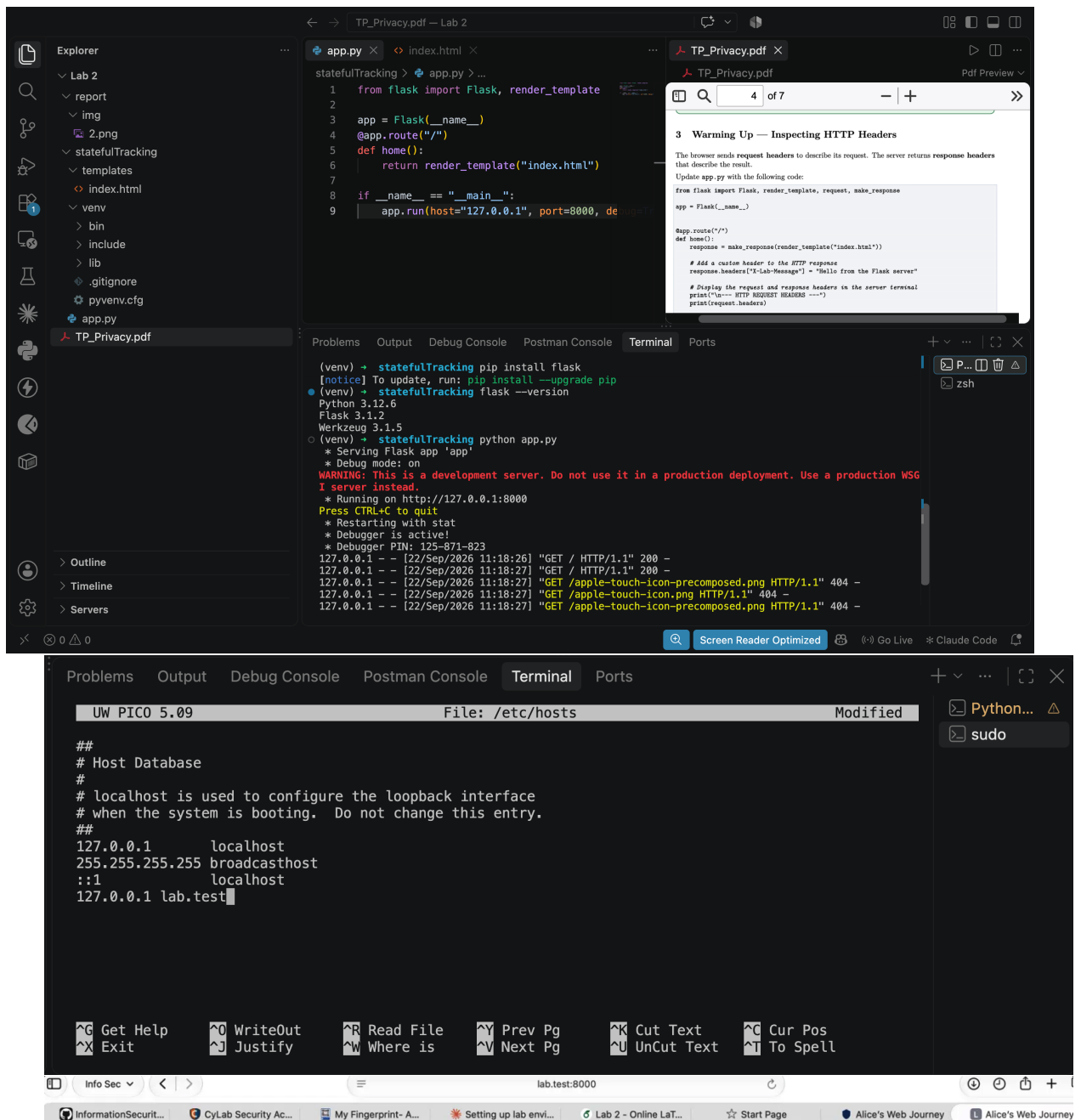
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September 22, 2026

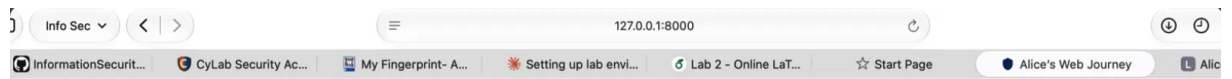
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1 Parts 1 and 2



Welcome to Alice's Web Journey



Welcome to Alice's Web Journey

2 Part 3

```
(venv) .-> statefulTracking python app.py
127.0.0.1 - - [22/Sep/2026 11:18:27] "GET /favicon.ico HTTP/1.1" 404 -
127.0.0.1 - - [22/Sep/2026 11:22:58] "GET / HTTP/1.1" 200 -
* Detected change in /Users/radradar/Documents/CS_2/Semester_7/Information Security/Lab 2/statefulTracking/app.py', reloading
* Restarting with stat
* Debugger is active!
* Debugger PIN: 123-871-823

---- HTTP REQUEST HEADERS ----
Host: lab.test:8000
User-Agent: Mozilla/5.0 (Macintosh; Intel Mac OS X 10_15_7) AppleWebKit/605.1.15 (KHTML, like Gecko) Version/27.0 Safari/605.1.15
Upgrade-Insecure-Requests: 1
Accept: text/html,application/xhtml+xml,application/xml;q=0.9,*/*;q=0.8
Accept-Language: en-US,en;q=0.9
Priority: u=0, i
Accept-Encoding: gzip, deflate
Connection: keep-alive

---- HTTP RESPONSE HEADERS ----
Content-Type: text/html; charset=utf-8
Content-Length: 172
X-Lab-Message: Hello from the Flask server

127.0.0.1 - - [22/Sep/2026 11:29:27] "GET / HTTP/1.1" 200 -

---- HTTP REQUEST HEADERS ----
Host: lab.test:8000
User-Agent: Mozilla/5.0 (Macintosh; Intel Mac OS X 10_15_7) AppleWebKit/605.1.15 (KHTML, like Gecko) Version/27.0 Safari/605.1.15
Upgrade-Insecure-Requests: 1
Accept: text/html,application/xhtml+xml,application/xml;q=0.9,*/*;q=0.8
Accept-Language: en-US,en;q=0.9
Priority: u=0, i
Accept-Encoding: gzip, deflate
Connection: keep-alive

---- HTTP RESPONSE HEADERS ----
Content-Type: text/html; charset=utf-8
Content-Length: 172
X-Lab-Message: Hello from the Flask server

127.0.0.1 - - [22/Sep/2026 11:29:32] "GET / HTTP/1.1" 200 -

---- HTTP REQUEST HEADERS ----
Host: lab.test:8000
User-Agent: Mozilla/5.0 (Macintosh; Intel Mac OS X 10_15_7) AppleWebKit/605.1.15 (KHTML, like Gecko) Version/27.0 Safari/605.1.15
Upgrade-Insecure-Requests: 1
Accept: text/html,application/xhtml+xml,application/xml;q=0.9,*/*;q=0.8
Accept-Language: en-US,en;q=0.9
Priority: u=0, i
Accept-Encoding: gzip, deflate
Connection: keep-alive

---- HTTP RESPONSE HEADERS ----
Content-Type: text/html; charset=utf-8
Content-Length: 172
X-Lab-Message: Hello from the Flask server

127.0.0.1 - - [22/Sep/2026 11:29:42] "GET / HTTP/1.1" 200 -
```

Welcome to Alice's Web Journey

The screenshot shows the Chrome DevTools Network tab. The top toolbar includes buttons for 'Filter Full URL', 'All', 'Disable Caches', 'Import', and 'Export'. The main table has columns: Name, Domain, Type, Status, Initiator, Transfer Size, Time, and a timeline view. A single request is visible: 'lab.test' with domain 'lab.test', type 'document', status '200', and a transfer size of '390 B'. The timeline shows a single bar for this request.

Name	Domain	Type	Status	Initiator	Transfer Size	Time
lab.test	lab.test	document	200	—	390 B	5.02ms

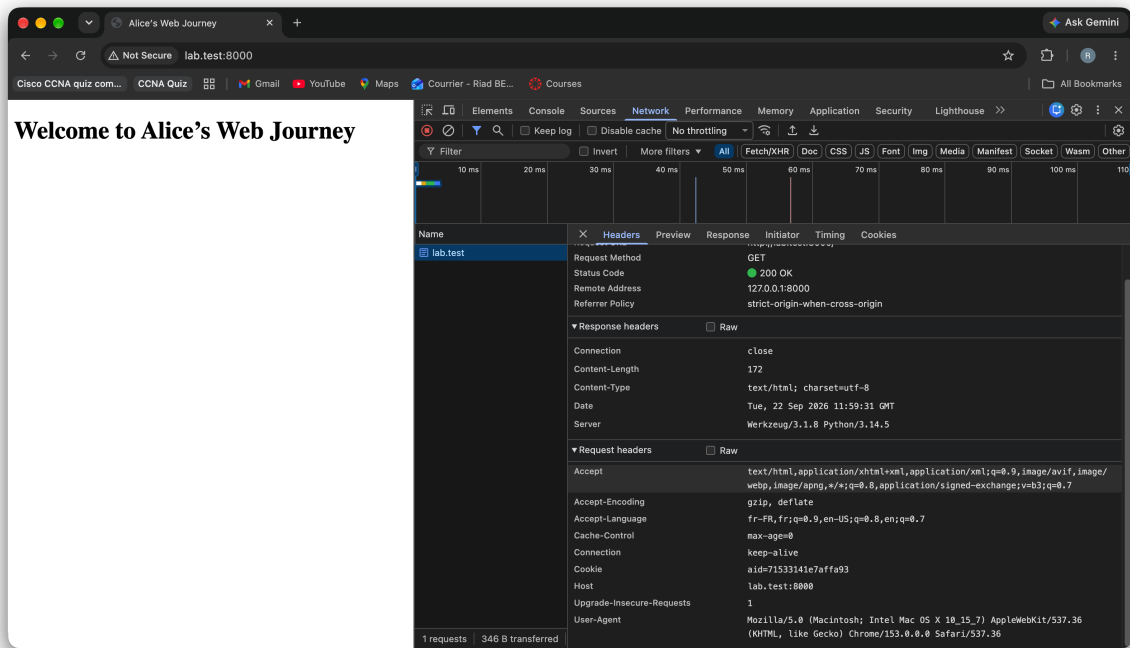
No cookie is exchanged yet, so the server cannot recognize a returning visitor

3 Part 4

The screenshot shows the Chrome DevTools Network tab with a single request selected. The 'Headers' panel is expanded, showing the following details:

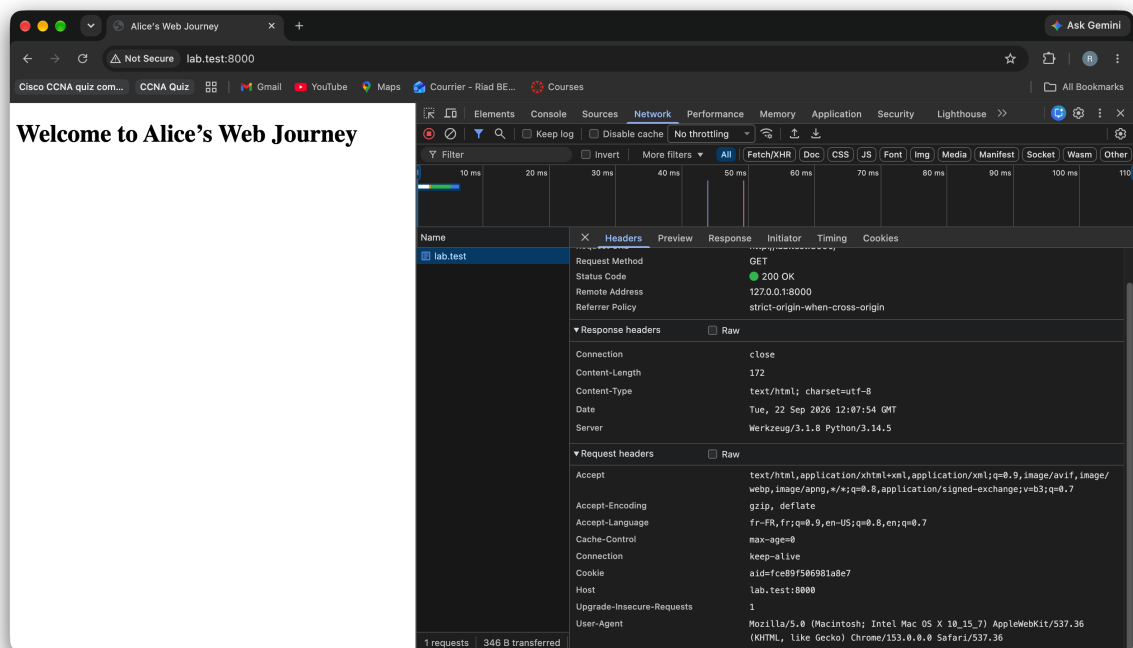
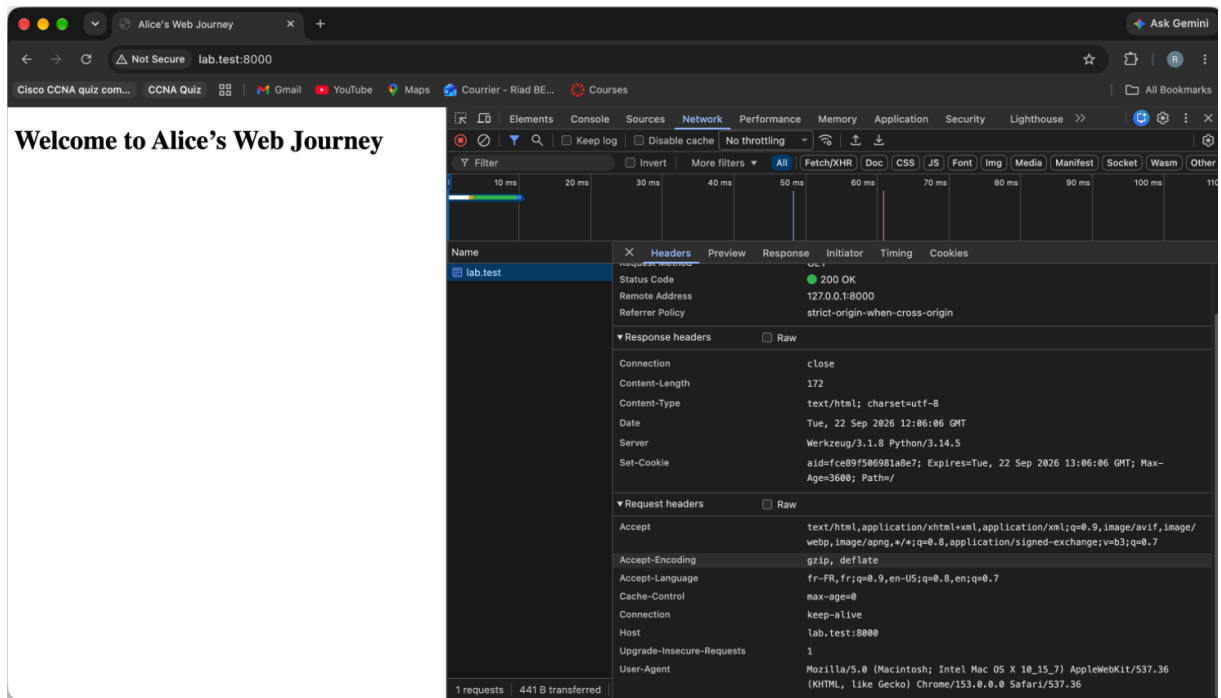
- Request:** Request URL: http://lab.test:8000/, Request Method: GET, Status Code: 200 OK, Remote Address: 127.0.0.1:8000, Referrer Policy: strict-origin-when-cross-origin.
- Response Headers:** Connection: close, Content-Length: 172, Content-Type: text/html; charset=utf-8, Date: Tue, 22 Sep 2026 11:52:53 GMT, Server: Werkzeug/3.1.8 Python/3.14.5, Set-Cookie: aid=71533141e7affa93; Path=/. (Note: The original image contains a typo 'aid=71533141e7affa93' which has been corrected to 'aid=71533141e7affa93' based on the context of the document.)
- Request Headers:** Accept: text/html,application/xhtml+xml,application/xml;q=0.9,image/avif,image/webp,image/apng,*/*;q=0.8,application/signed-exchange;v=b3;q=0.7, Accept-Encoding: gzip, deflate, Accept-Language: fr-FR,fr;q=0.9,en-US;q=0.8,en;q=0.7, Cache-Control: max-age=0, Connection: keep-alive, Host: lab.test:8000, Upgrade-Insecure-Requests: 1, User-Agent: Mozilla/5.0 (Macintosh; Intel Mac OS X 10_15_7) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/153.0.0.0 Safari/537.36.

On the first visit, the server sends **Set-Cookie: aid=...; Path=/. On the next visits, the browser sends it back in the Cookie header, so the server recognizes the same browser. The cookie is a session cookie (no expiration) for lab.test.**



Is the cookie still present after restarting the browser? Yes, with the same identifier, because Chrome was not fully quit.

```
is_new:
response.set_cookie(
    key="aid",
    value=aid,
    max_age=3600
)
return response
```

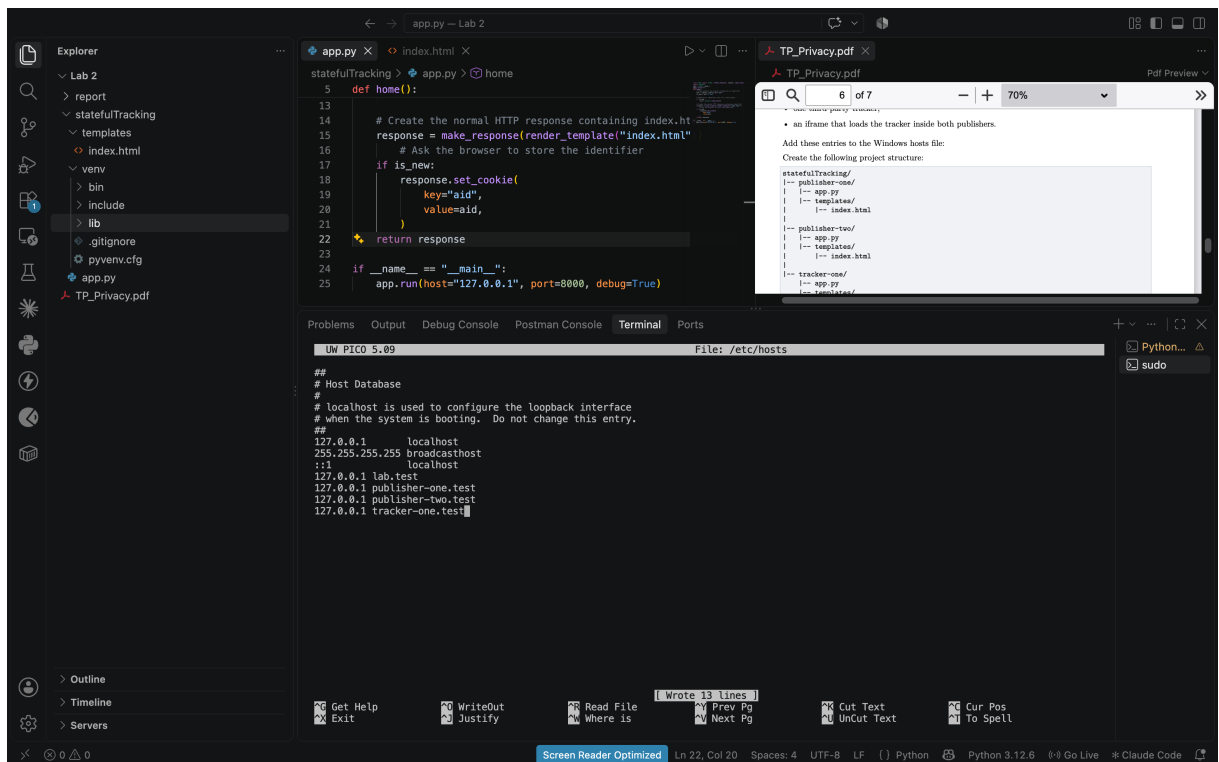


Attribute max_age: with max_age=3600, the server sends Expires and Max-Age=3600 in Set-Cookie. The cookie becomes persistent: it expires one hour after being set and survives a browser restart with the same identifier, independently of the session restore option.

4 Part 5

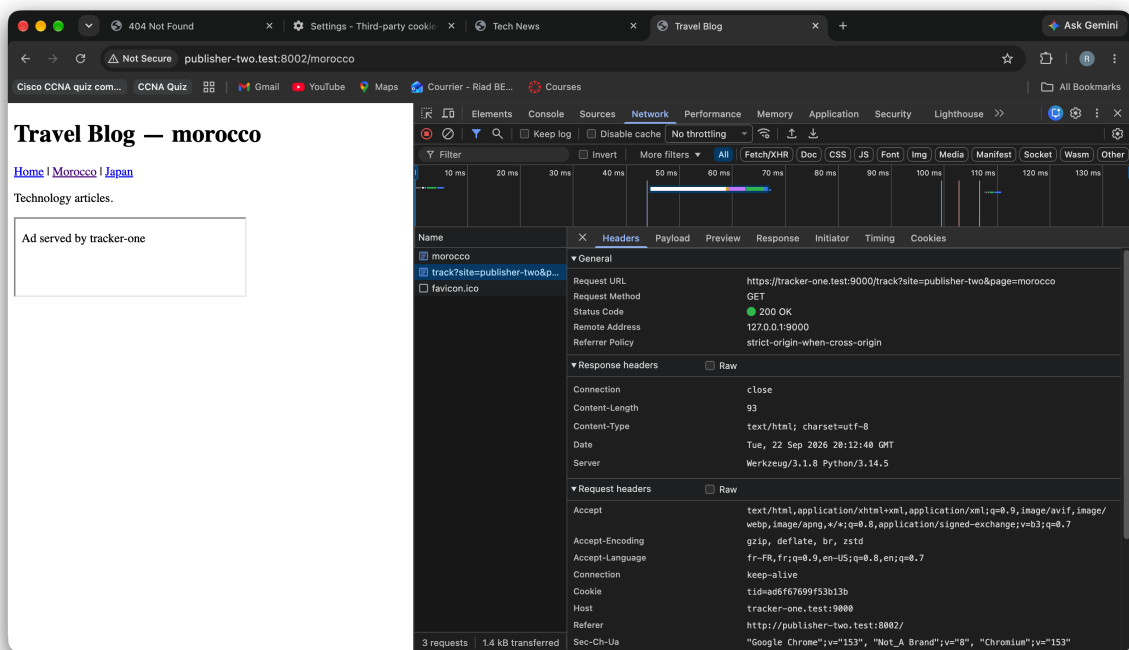
4.1 Setup:

```
<> tracker.html ×  app.py .../tracker-one ×  app.py statefulTracking ×
statefulTracking > tracker-one >  app.py > track
1  from flask import Flask, render_template, request, make_response
2  from datetime import datetime
3  import secrets
4
5  app = Flask(__name__)
6  log = []
7
8  @app.route("/track")
9  def track():
10     tid = request.cookies.get("tid")
11     is_new = tid is None
12     if is_new:
13         tid = secrets.token_hex(8)
14
15     log.append((datetime.now().strftime("%H:%M:%S"), tid,
16               request.args.get("site"), request.args.get("page")))
17
18     print(f"\n--- PROFILE {tid} ---")
19     for time, t, site, page in log:
20         if t == tid:
21             print(time, site, page)
22
23     response = make_response(render_template("tracker.html"))
24     if is_new:
25         response.set_cookie("tid", tid, max_age=365*24*3600,
26                             samesite="None", secure=True)
27     return response
28
29 if __name__ == "__main__":
30     app.run(host="127.0.0.1", port=9000, debug=True, ssl_context="adhoc")
```



For the rest of the code concerning the other pages of the challenge, please check the repo. All codes are pushed with this report.

4.2 Tracker in action:



The screenshot shows a VS Code editor with the following components:

- Explorer:** A file tree on the left showing a project structure with folders like 'report', 'statefulTracking', and 'templates', and files like 'app.py' and 'TP_Privacy.pdf'.
- Editor:** The main workspace shows a Python file 'app.py' with code for a Flask application. The code includes imports for 'datetime' and 'secrets', and defines a 'track' function that logs requests.
- Terminal:** A terminal window at the bottom displays the output of running the application. It shows the Flask server starting, the debugger being active, and a series of HTTP requests and responses. The logs include timestamps, IP addresses, and the specific requests being made to the application.
- PDF Viewer:** A PDF viewer on the right shows a document titled 'TP_Privacy.pdf'. It displays a 'Your Task' section with the instruction: 'Update the three applications so that: 1. both publisher websites run independently;'. The PDF is currently on page 6 of 7.

Each publisher embeds an `iframe` pointing to `tracker-one.test`, with the publisher name and page passed as query parameters. On the first request, the tracker generates a persistent identifier and stores it in a cookie.

Because the cookie belongs to `tracker-one.test`, the browser sends it back on every `iframe` request, whichever publisher is being visited. The tracker therefore receives the same identifier from both websites and reconstructs a single profile in chronological order:

```

Profil ad6f67699f53b13b
20:09:32 publisher-one home
20:09:47 publisher-two home
20:10:03 publisher-one ai
20:10:05 publisher-one security
20:10:08 publisher-two morocco

```

The two publishers never exchange data, but the shared third-party tracker links Alice's activity across both sites.

5 Part 6

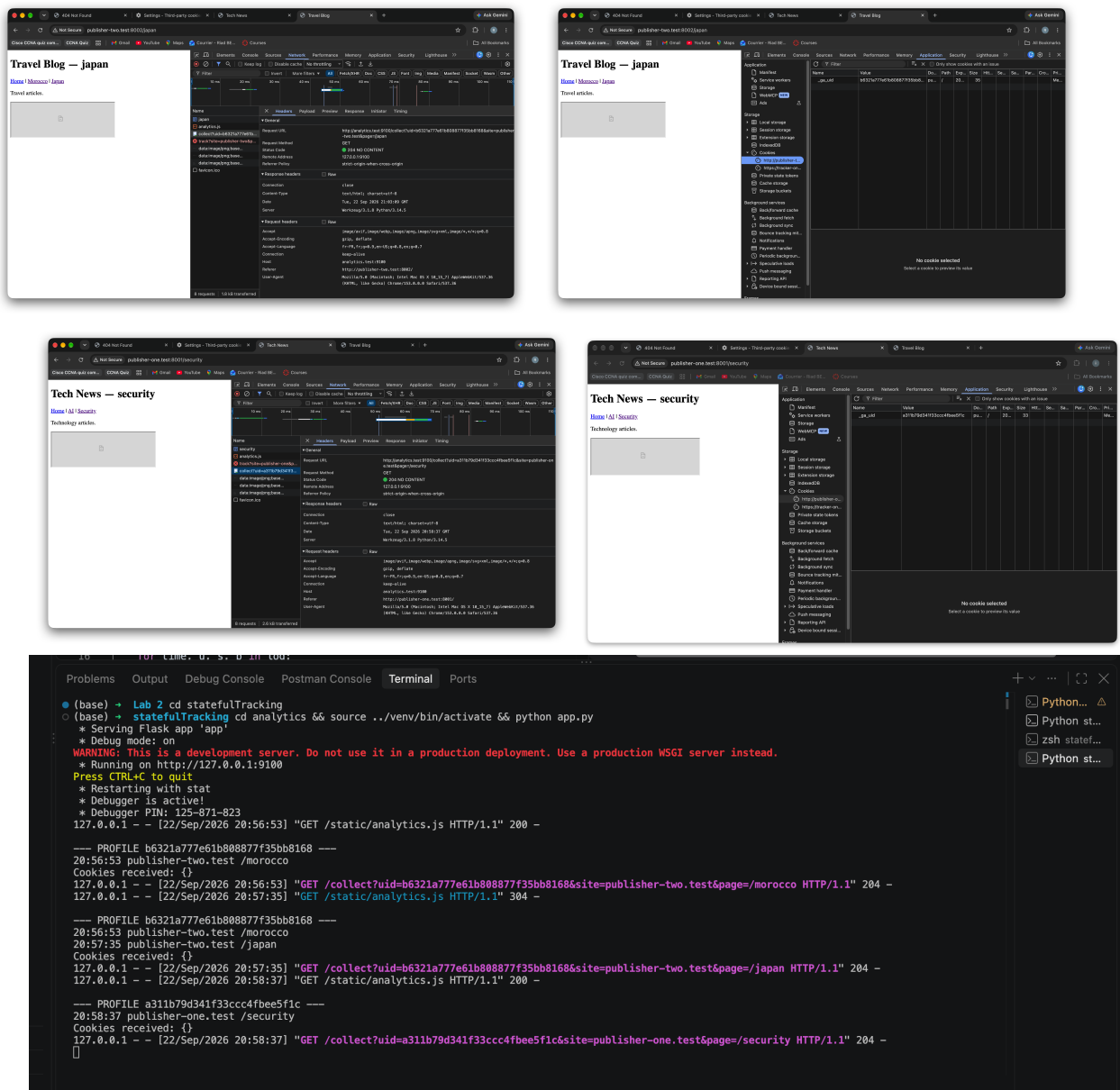
5.1 Setup

```
app.py × JS analytics.js × <> index.html ×
statefulTracking > analytics > app.py > ...
1 from flask import Flask, request
2 from datetime import datetime
3
4 app = Flask(__name__)
5 log = []
6
7 @app.route("/collect")
8 def collect():
9     uid = request.args.get("uid")
10    site = request.args.get("site")
11    page = request.args.get("page")
12
13    log.append((datetime.now().strftime("%H:%M:%S"), uid, site, page))
14
15    print(f"\n--- PROFILE {uid} ---")
16    for time, u, s, p in log:
17        if u == uid:
18            print(time, s, p)
19    print("Cookies received:", dict(request.cookies))
20
21    return "", 204
22
23 if __name__ == "__main__":
24     app.run(host="127.0.0.1", port=9100, debug=True)
```

```
app.py × JS analytics.js × <> index.html ×
statefulTracking > analytics > static > JS analytics.js > ...
1 function getCookie(name) {
2     const match = document.cookie.match(new RegExp("(^| )" + name + "=[^;]+)"));
3     return match ? match[2] : null;
4 }
5
6 let uid = getCookie("_ga_uid");
7
8 if (!uid) {
9     uid = Math.random().toString(16).slice(2) + Math.random().toString(16).slice(2);
10    // 4. keep it for future visits
11    document.cookie = "_ga_uid=" + uid + "; path=/; max-age=" + 365 * 24 * 3600;
12 }
13
14 // send browsing activity to the analytics server
15 const img = new Image();
16 img.src = "http://analytics.test:9100/collect"
17         + "?uid=" + uid
18         + "&site=" + location.hostname
19         + "&page=" + location.pathname;
```

For the rest of the code concerning the other pages of the challenge, please check the repo. All codes are pushed with this report.

5.2 Analytics in action



The analytics script is downloaded from `analytics.test` but executes in the publisher's page, so the identifier it creates with `document.cookie` is a first-party cookie of the publisher: `_ga_uid` appears under `publisher-one.test` and `publisher-two.test`, not under `analytics.test`.

Same identifier on both publishers? No (`a311b79d341f33ccc4fbee5f1c` and `b6321a777e61b808877f35bb8168`), because cookies are isolated by domain: the script on one publisher cannot read the other's cookie.

Does the analytics server receive the first-party cookies? No. The collect requests contain no Cookie header, since a cookie is only sent to its own domain. The identifier reaches the server only because the script copies it into the URL parameters.

Profiles. The service builds two separate profiles, one per publisher, instead of a single cross-site profile as in Challenge 1.

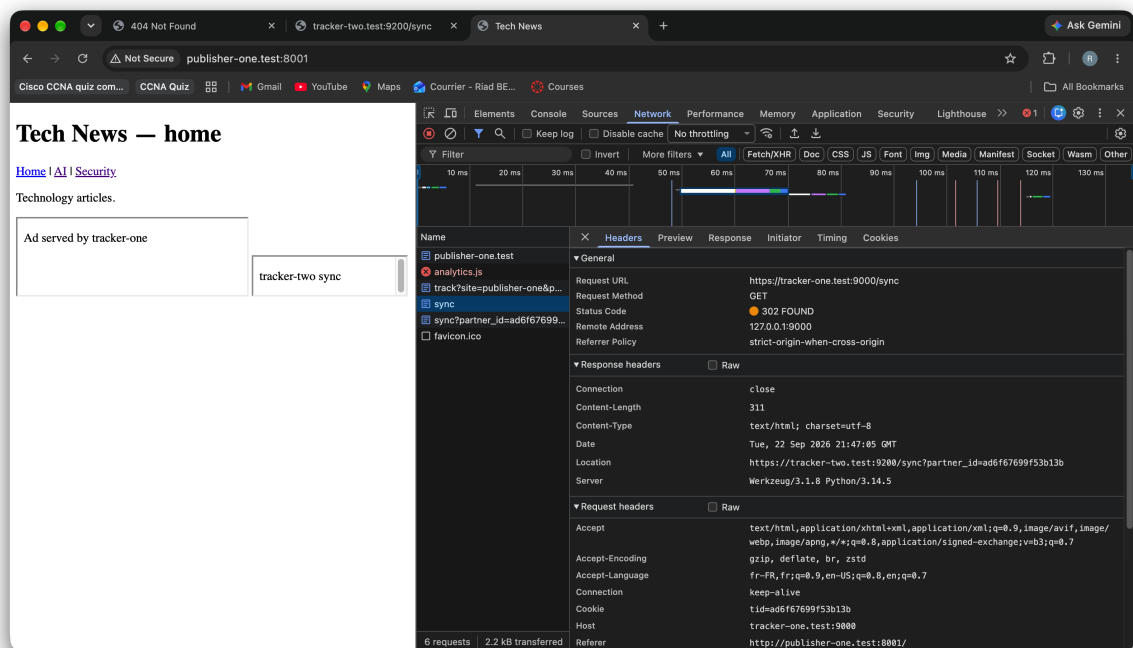
6 Part 7

For setup code of the challenge, please check the repo. All codes are pushed with this report.

```
Problems Output Debug Console Postman Console Terminal Ports
(base) → Lab 2 cd statefulTracking
(base) → statefulTracking cd tracker-two && source ../venv/bin/activate && python app.py
* Serving Flask app 'app'
* Debug mode: on
WARNING: This is a development server. Do not use it in a production deployment. Use a production WSGI server instead.
* Running on https://127.0.0.1:9200
Press CTRL+C to quit
* Restarting with stat
* Debugger is active!
* Debugger PIN: 125-871-823

[TRACKER-TWO] my id: ee123ae4cb7ebe39, partner id: None
--- ID MAPPING TABLE ---
tracker-one None = tracker-two ee123ae4cb7ebe39
127.0.0.1 - - [22/Sep/2026 21:41:41] "GET /sync HTTP/1.1" 200 -
127.0.0.1 - - [22/Sep/2026 21:41:41] "GET /favicon.ico HTTP/1.1" 404 -

[TRACKER-TWO] my id: ee123ae4cb7ebe39, partner id: ad6f67699f53b13b
--- ID MAPPING TABLE ---
tracker-one None = tracker-two ee123ae4cb7ebe39
tracker-one ad6f67699f53b13b = tracker-two ee123ae4cb7ebe39
127.0.0.1 - - [22/Sep/2026 21:47:05] "GET /sync?partner_id=ad6f67699f53b13b HTTP/1.1" 200 -
█
```



Each tracker has its own cookie on its own domain and cannot read the other's. **tracker-one** answers with a redirect to **tracker-two.test/sync?partner_id=<its own id>**. Following it, the browser contacts **tracker-two.test** and automatically attaches tracker two's cookie, so tracker two receives both identifiers at once and stores the pair:

tracker-one ad6f67699f53b13b ; **tracker-two** ee123ae4cb7ebe39